Project Proposal: Reproducing OWL and Testing Whether Outlier Weighed Layerwise Sparsity Still Helps Small Modern LLMs

Tuan Khang Phan and Xuan Gia Han Nguyen
University of South Florida

1 Introduction

We will reproduce "Outlier Weighed Layerwise Sparsity (OWL): A Missing Secret Sauce for Pruning LLMs to High Sparsity" (Yin et al., 2024). The paper studies one-shot unstructured pruning of LLMs with no retraining. Strong methods such as SparseGPT (Frantar and Alistarh, 2023) and Wanda (Sun et al., 2024) prune every layer to the same sparsity. OWL questions this choice and links pruning quality to outlier features, a few hidden dimensions with very large activations (Kovaleva et al., 2021; Dettmers et al., 2022).

OWL scores each weight with the Wanda metric $A_{ij} = \|\mathbf{X}_j\|_2 \cdot |W_{ij}|$. For each Transformer block l it computes the Layerwise Outlier Distribution (LOD) D^l, the fraction of weights whose score is over M times the block mean. Blocks with more outliers are pruned less: $S_l \propto 1 - D^l$, rescaled into a band of width 2λ whose mean is the target S.

We target four claims. (C1) OWL improves Wanda and SparseGPT at high sparsity. At 70% on LLaMA-7B, WikiText perplexity drops from 85.77 to 24.55 with Wanda and from 26.30 to 19.49 with SparseGPT, although OWL hurts SparseGPT on OPT-6.7B (20.29 to 22.48). (C2) With Wanda, the gain grows as models shrink: 61.22 points at 7B but only 6.62 at 65B. (C3) Zero-shot accuracy also improves on seven tasks. (C4) The LOD of dense LLaMA models is far from uniform and roughly U-shaped, and good pruning methods keep outliers.

2 Reproducing Main Results

We reproduce C1, C3, C4 and the 7B and 13B part of C2. We skip the 30B and 65B models, the CPU speedup study, LoRA fine-tuning and the appendix studies.

Models. Our main model is LLaMA-7B (Touvron et al., 2023). The README's checkpoint was removed from Hugging Face, so we use `huggyllama/llama-7b`. We also run OPT-6.7B (Zhang et al., 2022) and LLaMA-13B, with the paper's hyperparameters (M=5, λ=0.08 for 7B, M=7 for 13B, M=10 for OPT).

Data. Calibration uses 128 random 2048-token sequences from the first C4 shard (Raffel et al., 2020). Perplexity uses 2048-token windows of the WikiText-2 (Merity et al., 2017) test split, which the paper calls validation. Zero-shot accuracy covers BoolQ (Clark et al., 2019), RTE (Wang et al., 2018), HellaSwag (Zellers et al., 2019), WinoGrande (Sakaguchi et al., 2021), ARC-e, ARC-c (Clark et al., 2018) and OpenBookQA (Mihaylov et al., 2018). For R3 we run the authors' own zero-shot scripts (lm-eval-harness 0.3.0, Gao et al., 2024) on the saved pruned model. Table 4 does not say whether it reports acc or acc_norm, so we log both.

Experiments. (R1) Table 3: magnitude pruning, plus Wanda and SparseGPT each with uniform and OWL sparsity, at 70% on all three models, including the OPT-6.7B case where OWL hurts. (R2) Figure 2 and Table 8: a 40% to 80% sweep on LLaMA-7B with Wanda, comparing OWL with uniform, global, ER, ER-plus and OWL-inverse allocations. (R3) Table 4: zero-shot accuracy of the dense and 70% sparse LLaMA-7B models. (R4) Table 1 and Figure 1: the LOD of each dense model, recomputed after pruning with the dense model's threshold.

Success criterion. We first check the dense LLaMA-7B perplexity of 5.68 and report our own baselines next to the paper's. A claim counts as reproduced if every R1 setting shows the paper's direction of OWL gain or loss and our 70% perplexities are within 15% of the paper's. Methods within 2% count as tied.

3 Plans for Extension

C2 was only tested on LLaMA-1 (7B to 65B), trained on at most 1.4T tokens. Modern small models see far more data, about 15T tokens for Llama 3 (Grattafiori et al., 2024) and 18T for Qwen2.5 (Qwen Team et al., 2024). Evidence for OWL on them is mixed: with SparseGPT at 70%, it nearly halves Llama-3-8B perplexity, from 85.84 to 48.07 (Sieberling et al., 2025). But OWL slightly hurts a 1B Llama 3 model at 50% sparsity in a multilingual setting (Choenni and Titov, 2025), and the layer-wise allocation of Jamshidi and Shvets (2026) gives only marginal gains on Llama-3-8B. ATP (Huang et al., 2025) even argues for a schedule that simply increases with depth. Systematic tests across sizes within one modern family, especially below 3B, remain limited. We ask whether OWL's benefit still grows as models shrink, and whether it comes from aligning sparsity with outlier-rich blocks.

E1: Scaling within a family. We prune Qwen2.5 0.5B, 1.5B, 3B and 7B, and Llama-3.2-1B, Llama-3.2-3B and Llama-3.1-8B, at 50%, 60% and 70% with Wanda and SparseGPT, each with uniform and OWL allocation, using the paper's 7B setting (M=5, λ=0.08). Small models start from higher perplexity, so besides the absolute drop we report the OWL gain in log perplexity, $\Delta L = \log \mathrm{PPL}_{\mathrm{uni}} - \log \mathrm{PPL}_{\mathrm{OWL}}$, separately per method, plus the loss relative to dense and mean zero-shot accuracy at 70%. Size also changes width and depth, so this is a trend, not a causal effect. All blocks in a model are equal in size, so block sparsities average to the global one, which we also measure.

Controls. To test whether the position of each block's sparsity matters, we randomly permute OWL's block sparsities (three permutations per model, Wanda, 70%), keeping the values but breaking their link to the LOD. Two extra calibration seeds for Qwen2.5-0.5B and Llama-3.2-1B check the sensitivity to calibration samples.

E2: Does the LOD predict the gain? We compute the LOD of all ten dense models (three LLaMA-1/OPT, seven modern). We measure unevenness with the coefficient of variation across blocks and the U shape with U, the mean LOD of the outer 20% of blocks minus that of the middle 60%. With ten related models, Spearman correlations with ΔL (per method, with bootstrap intervals, leaving out one family at a time) are exploratory.

Exp.	Models	Runs	A100 h
R1	LLaMA-7B/13B, OPT-6.7B	15	3
R2	LLaMA-7B	28	2
R3	LLaMA-7B	6	3
R4, E2	10 dense models	10	1
E1	7 Qwen2.5 / Llama-3	113	15
E3	2 small models	40	2
E4	Qwen2.5-1.5B/7B	16	2
Total		228	28

Table 1: Planned runs (pruning plus evaluation, excluding dense baselines) and estimated A100 hours.

E3: Hyperparameter transfer. A weak result could just mean bad settings. We run the paper's grid ($M \in \{3, 5, 7, 10\}$, $\lambda \in \{0.02, 0.05, 0.08, 0.1, 0.2\}$) on Qwen2.5-1.5B and Llama-3.2-1B at 70% with Wanda, select on the C4 validation shard (disjoint from the calibration shard), apply the choice to the whole family, and report default and tuned OWL.

E4 (optional): Calibration language. We calibrate Qwen2.5-1.5B and 7B on Vietnamese Wikipedia instead of C4 (Kurz et al., 2026) and compare LOD profiles and English and Vietnamese perplexity.

4 Resources

No paid inference API is needed: all models are open-weight on Hugging Face.

Memory. A 7B model peaks at about 18 GB in fp16 (13.5 GB of weights plus one block's calibration activations), so 7B and 8B models need a 24 GB GPU, LLaMA-13B about 31 GB, and models up to 3B fit in 16 GB.

Compute. The paper reports 266 s for SparseGPT on a 7B model and under 1 s for the Wanda metric. Table 1 estimates 28 A100 hours, to be confirmed by profiling a small Wanda and a 7B SparseGPT run. Zero-shot evaluation and reruns may cost more, so we budget 60. We use USF's GAIVI cluster (15 A100, 6 H100, 10 L40S), free with a supervisor's approval. Preempted runs are resubmitted; finished ones are skipped.

Backup. Kaggle's 16 GB T4s (30 hours a week) cover models up to 3B, Modal's $30 monthly credit buys 14 A100 hours, and 60 rented hours ($0.3 to $1.8 each) cost under $110.

References

Rochelle Choenni and Ivan Titov. 2025. M-Wanda: Improving one-shot pruning for multilingual LLMs. *arXiv preprint arXiv:2505.21171*.

Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. 2019. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*.

Tim Dettmers, Mike Lewis, Younes Belkada, and Luke Zettlemoyer. 2022. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems*, volume 35.

Elias Frantar and Dan Alistarh. 2023. SparseGPT: Massive language models can be accurately pruned in one-shot. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*. PMLR.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac'h, et al. 2024. The language model evaluation harness.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, et al. 2024. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Weizhong Huang, Yuxin Zhang, Xiawu Zheng, Fei Chao, and Rongrong Ji. 2025. Determining layer-wise sparsity for large language models through a theoretical perspective. *arXiv preprint arXiv:2502.14770*.

Yazdan Jamshidi and Alexey Shvets. 2026. PALS: Percentile-aware layerwise sparsity for LLM pruning. *arXiv preprint arXiv:2607.07557*.

Olga Kovaleva, Saurabh Kulshreshtha, Anna Rogers, and Anna Rumshisky. 2021. BERT busters: Outlier dimensions that disrupt transformers. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*.

Simon Kurz, Jian-Jia Chen, Lucie Flek, and Zhixue Zhao. 2026. On the limitations of language-targeted pruning: Investigating the calibration language impact in multilingual LLM pruning. *Transactions of the Association for Computational Linguistics*, 14.

Stephen Merity, Caiming Xiong, James Bradbury, and Richard Socher. 2017. Pointer sentinel mixture models. In *International Conference on Learning Representations*.

Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. 2018. Can a suit of armor conduct electricity? a new dataset for open book question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*.

Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, et al. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140).

Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2021. WinoGrande: An adversarial winograd schema challenge at scale. *Communications of the ACM*, 64(9).

Oliver Sieberling, Denis Kuznedelev, Eldar Kurtic, and Dan Alistarh. 2025. EvoPress: Accurate dynamic model compression via evolutionary search. In *Proceedings of the 42nd International Conference on Machine Learning*.

Mingjie Sun, Zhuang Liu, Anna Bair, and J. Zico Kolter. 2024. A simple and effective pruning approach for large language models. In *The Twelfth International Conference on Learning Representations*.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.

Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2018. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP*.

Lu Yin, You Wu, Zhenyu Zhang, Cheng-Yu Hsieh, Yaqing Wang, Yiling Jia, Gen Li, Ajay Jaiswal, Mykola Pechenizkiy, Yi Liang, Michael Bendersky, Zhangyang Wang, and Shiwei Liu. 2024. Outlier weighed layerwise sparsity (OWL): A missing secret sauce for pruning LLMs to high sparsity. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. 2022. OPT: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*.